



Technische
Universität
Braunschweig

**Decision
Support**

Institut für Wirtschaftsinformatik

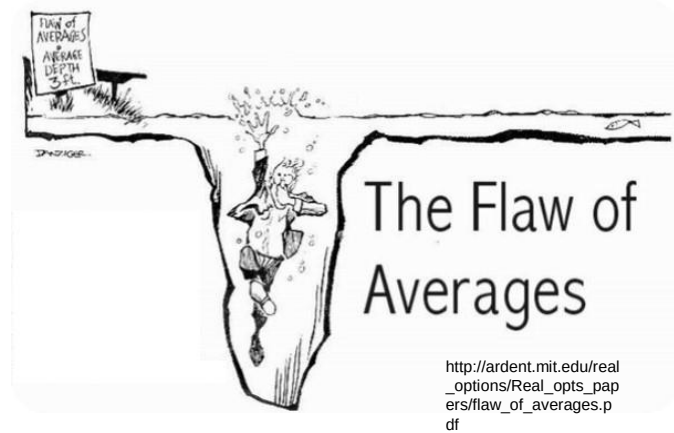
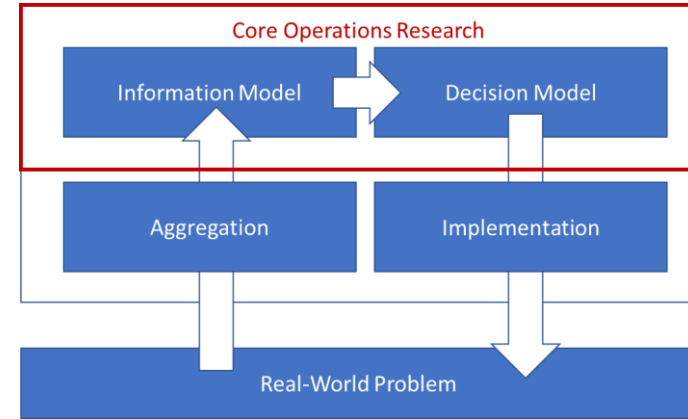


Data Driven Decision Making

Lecture 3 – Uncertainty

Recap & Today

- Full process of business analytics
- Information and decision model
- Data comes first
- Finding the “right” information model is challenging
- Decision making complexity increases with more elaborate information model.
- Uncertainty in the information model
- How can we model uncertainty?
- How can we make decisions under uncertainty?
- Case Study: Delivery routing with uncertain travel times



Agenda

- **Uncertainty**
- Modeling Uncertainty
- Decision Making under Uncertainty
- Case Study: Robust Vehicle Routing in Urban Logistics

Introduction

Examples:

- Stock market
- Demand
- Supply
- Innovation
- Politics



Differ in

- Frequency
- Range
- Disruptivity
- Predictability



playstation.com

Uncertainty in Mobility and Logistics

- Customers: requests, demand volumes, service times, availability, behavior
- Drivers: availability, behavior, range, skills
- Environment: travel times, parking, network availability



Uncertainty in Travel Times

- Weather
 - Continuous
 - Predictable
- Traffic
 - Systemic / follows observable patterns
 - Spontaneous (accidents, congestion)
 - Partially predictable
- Traffic Management
 - Discrete Events
 - Subjected to framework conditions and previous events
 - Partially predictable



Possible Consequences

- Complete cancellation of planned activities
 - Transport: Package is not delivered
 - Mobility: Workplace is inaccessible
 - Disruption of planned activities
 - Transport: Delayed package delivery
→ Complete failure possible if customer is not present anymore
 - Mobility: Workplace is reached with delay
 - General undesirable effects (e.g.)
 - Resulting additional costs
 - Decrease of customer satisfaction
 - Successive activities are affected
- Consideration of uncertainty in decision making is crucial

Agenda

- Uncertainty
- **Modeling Uncertainty**
- Decision Making under Uncertainty
- Case Study: Robust Vehicle Routing in Urban Logistics

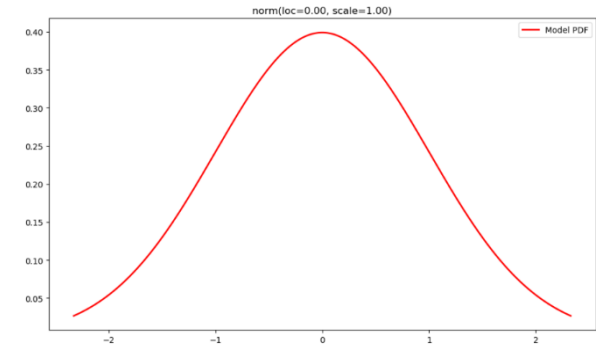
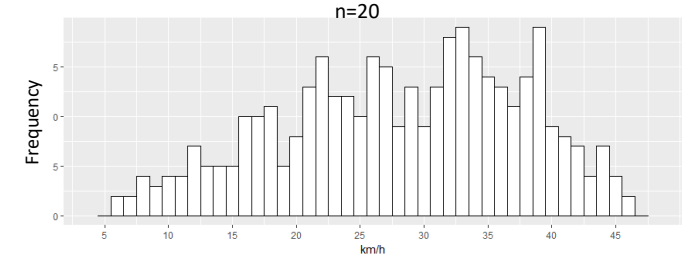
Modeling Uncertainty

Choice of model depends on level of information

1. Deterministic: All information is known
2. Stochastic: Probability of different information realizations is known
3. Quasi-stochastic: Probability of information realization is unknown

Basics

- Probability distribution
- Discrete or continuous
- Random Variable: X with realizations $x_1, \dots, x_i$ e.g., $x_1 = 20 \text{ km/h}$
- Probabilities $p_1, \dots, p_i$ of the realization of the corresponding x_i
- Expected value : $\mu = E[X] = \sum_i p_i x_i$ (or an integral)
- Variance: $\sigma^2 = E[(X - \mu)^2]$
- Standard deviation: $\sigma = \sqrt{\sigma^2}$
- Coefficient of variation: $CV = \frac{\sigma}{\mu}$
- Median and quantiles



- Metrics can be incorporated into decision making process
- Sampling: Generation of new observations from distribution

Determining a Probability Distribution

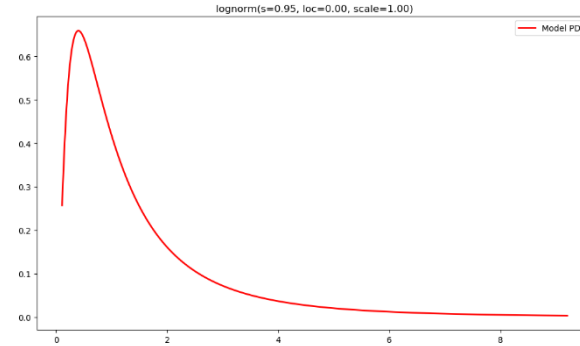
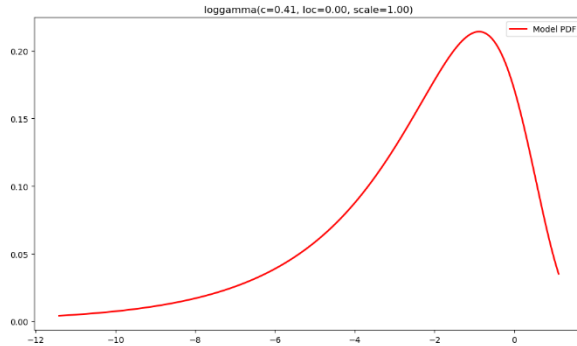
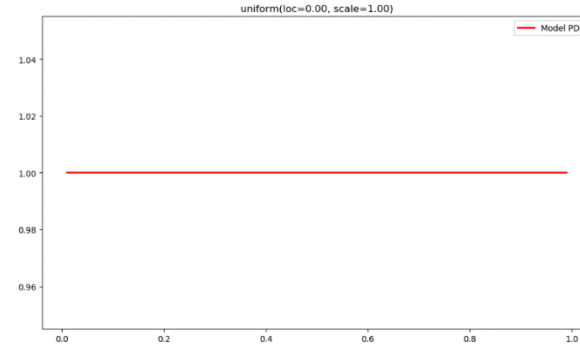
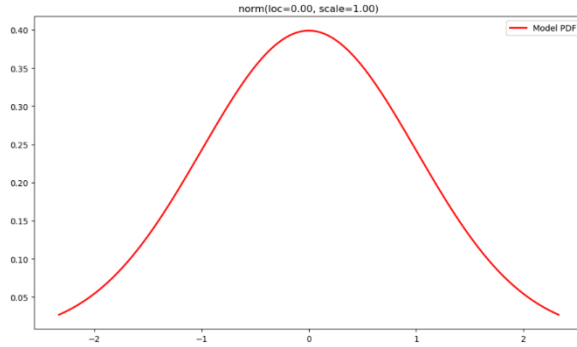
Simple assumption of a theoretical distribution and its corresponding parameters

- Uniform distribution
- Normal distribution
- Exponential distribution

Choice and adjustment of a probability distribution to existing data („Fitting“)

1. Choice of appropriate theoretical probability distribution
2. Estimation of corresponding parameters
3. Testing goodness-of-fit
 - Visualization
 - Goodness-of-fit statistics (e.g., Kolmogorov-Smirnov test)
4. Testing the appropriateness of estimated parameters (“bootstrapping”)

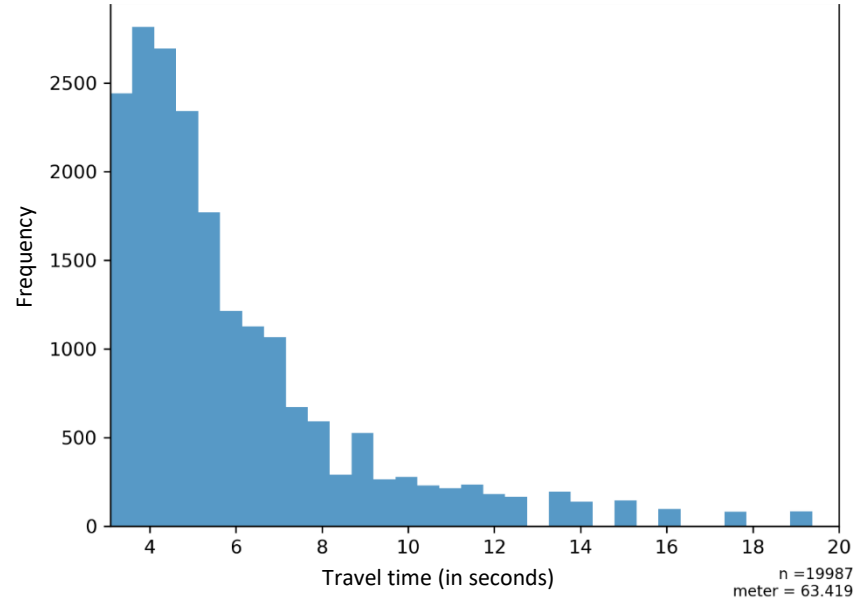
Theoretical Probability Density Functions



Example: Urban Travel Time Distribution I



Travel time observations for one street section



Fitting using Software

- Supporting: Python + Libraries



- Automated: EasyFit

EasyFit (Evaluierungsversion) - kmh-bsp - [Fit1]

Datei Bearbeiten Ansicht Analyse Optionen Extras Fenster Hilfe

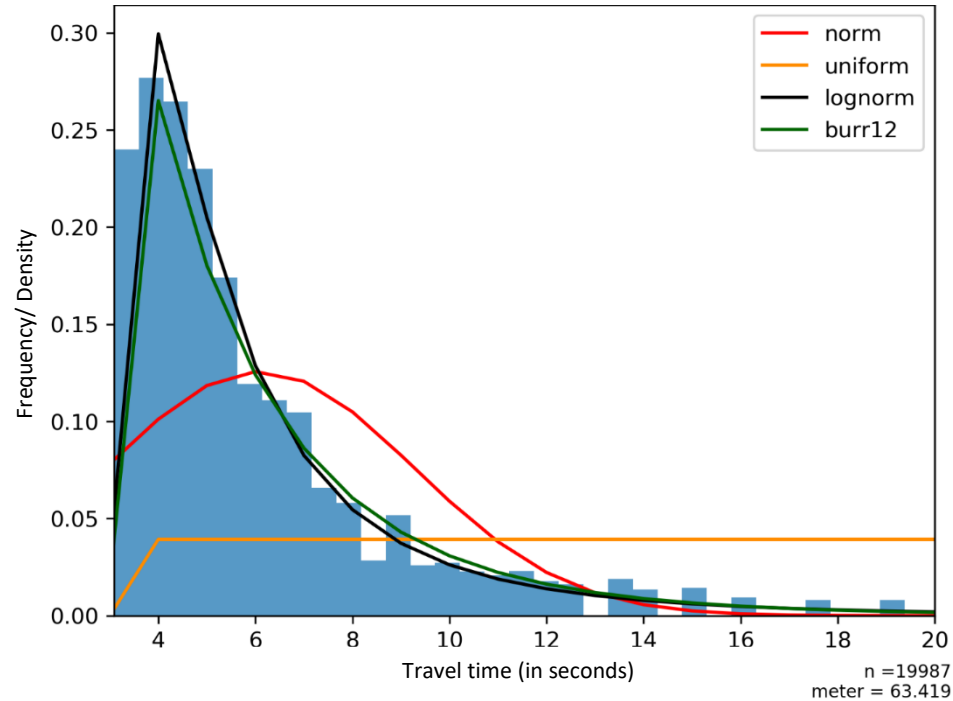
Projektstruktur

Graphen Zusammenfassung Anpassungsgüte

Anpassungstests - Zusammenfassung

#	Verteilung	Kolmogorow Smirnow		Anderson Darling		Chi-Quadrat	
		Statistik	Rang	Statistik	Rang	Statistik	Rang
1	Beta	0,02912	4	0,30016	3	7,6998	7
2	Burr	0,06428	18	2,3418	19	12,298	16
3	Burr (4P)	0,04197	8	1,4051	11	12,446	17
4	Cauchy	0,11929	42	8,7454	35	32,695	33
5	Chi-Squared	0,13206	49	22,213	47	83,546	45
6	Chi-Squared (2P)	0,10882	38	6,0575	28	19,716	26
7	Dagum	0,04834	14	0,8719	8	5,3982	2
8	Dagum (4P)	0,03489	5	0,59359	5	6,0943	3
9	Erlang	0,20776	51	31,725	49	92,871	47

Example: Urban Travel Time Distribution II



Possible Issues with Stochastic Modeling

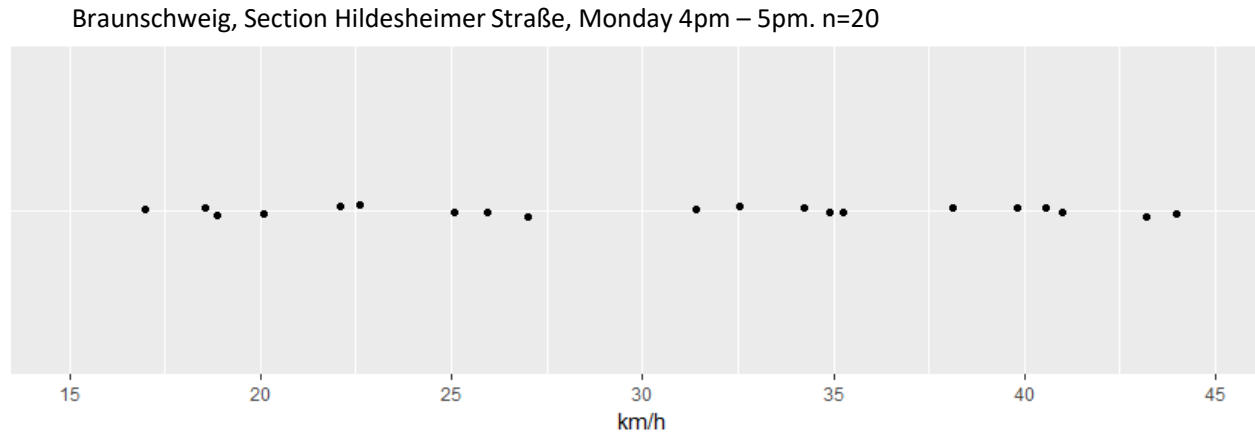
- Unavailability of data (e.g., cannot be observed a priori)
 - Lack of data recording
 - Change of system: Construction of a street
 - New concept (e.g., autonomous cars)
- Limited data available
 - Only few observations are available which leads to coarse estimation of the underlying distribution
- Assumption that path travel times are computed by convolution of independent distributions generated for street segments. Correlation is neglected.
 - Indeed, correlation of neighboring street segments is essential

Quasi-stochastic Modeling

- Based on scenarios
 - Create n scenario $s_n \in S$ (all scenarios)
 - Finite amount of possible realizations
 - „What-If“ analysis
 - e.g., traffic congestion: 10 km/h, no traffic congestion 50 km/h
 - Based on intervals $[l, u]$ (as range of possible realizations)
 - Smallest possible value l , highest possible value u
 - Possible realizations can have arbitrary values within interval limits
 - E.g., 20 km/h for $[10, 50]$ km/h
 - Both scenarios and intervals can be derived from domain knowledge and / or from data
- Depict an estimation of uncertainty without directly modeling a probability distribution

Constructing Scenarios from Data I

Example: Single observations of travel times

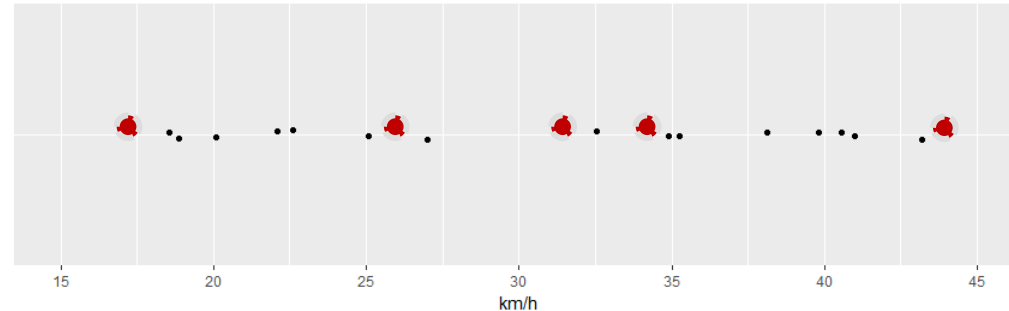


Constructing Scenarios from Data II

Possible approaches

- Set number of scenarios to construct (e.g., 5)
- Choose extreme scenarios (best-/worst-case)
- Choose in-between scenarios

Braunschweig, Section Hildesheimer Straße, Monday 4pm – 5pm. n=20



- Choice based on domain knowledge
 - Increasing the number of scenarios lead to more accurate estimations, in general
- Increases complexity of decision making

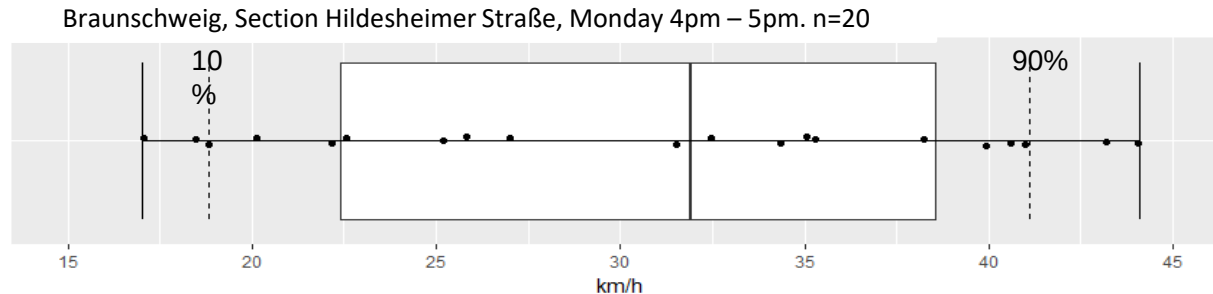
Determining Intervals from Data

Limits

- Upper: 90% Percentile
- Lower: 10% Percentile

Adjustment of limits

- Pessimistic: Shift towards worst-case
- Optimistic: Shift towards best-case
- Decreasing-/increasing interval span



Agenda

- Uncertainty
- Modeling Uncertainty (Descriptive and Predictive Analytics)
- **Decision Making under Uncertainty (Prescriptive Analytics)**
- Case Study: Robust Vehicle Routing in Urban Logistics

Example Knapsack Problem I

Example

A burglar with a knapsack gets into a house and wants to steal objects to sell later on.

- Each object has a value and a weight
- Total weight of a combination of objects has to fit the capacity of the knapsack
- Find combination of these objects with highest sum of selling prices.

Deterministic environment

- He knows exact values (v_i) and weights (w_i) of possible objects
- He knows the exact limit of this knapsack (W)

$$\begin{aligned} \max \quad & \sum_{i=1}^n v_i \cdot x_i \\ \text{s.t.} \quad & \sum_{i=1}^n w_i \cdot x_i \leq W \\ & x_i \in \{0,1\} \text{ for } i \in \{1, \dots, n\} \end{aligned}$$



Example Knapsack Problem II

Stochastic environment

- Possible uncertainty in the objective:
 - He does not know the exact selling prices on the black market.
- Possible uncertainty in the constraints:
 - He does not know the exact weights of the objects
 - He does not know the exact limit of his knapsack

Example

- **Random variable** in the objective
 - $\max \sum_{i=1}^n \mathbf{v}_i \cdot x_i$
- **Random variable** on LHS of constraint:
 - $\sum_{i=1}^n \mathbf{w}_i \cdot x_i \leq W$
- **Random variable** on RHS of constraint:
 - $\sum_{i=1}^n w_i \cdot x_i \leq \mathbf{W}$

Example

- **Random variable** in the objective
 - $\max \sum_{i=1}^n \mathbf{v}_i \cdot x_i$
- **Random variable** on LHS of constraint:
 - $\sum_{i=1}^n \mathbf{w}_i \cdot x_i \leq W$
- **Random variable** on RHS of constraint:
 - $\sum_{i=1}^n w_i \cdot x_i \leq \mathbf{W}$

Decision Making under Uncertainty

- Stochastic decision making differs from quasi-stochastic decision making
- Uncertainty can affect objective and / or constraints
- Realizations manifest after decisions are made
- Stochastic decision making:
Expected value and deviation is known
- Quasi-stochastic decision making:
Expected value and deviation is unknown



Stochastic Decision Making

- $\mathcal{X}$ domain of all feasible decision
- x a specific decision
- ξ random information (available after decision is made)
- F cost function to be minimized, denoted as $F(x, \xi)$
- Optimizing $F(x, \xi)$ directly is not possible
→ minimize expected value $\mathbb{E}[F(x, \xi)]$
- Optimal value: $\zeta^* = \min_{x \in \mathcal{X}} \{f(x) = \mathbb{E}[F(x, \xi)]\}$
- Set of optimal decisions: $\mathcal{S}^* = \{x \in \mathcal{X} : f(x) = \zeta^*\}$

Stochastic Decision Making II



Monte Carlo Sampling

- Choose n scenarios $(\xi)_{i=1}^n$ of ξ
- Approximate objective function by

$$\mathbb{E}[F(x, \xi)] \approx \frac{1}{n} \sum_{i=1}^n F(x, \xi_i)$$

- Solve deterministic right hand side
- Approximate optimal value:

$$\zeta_n^* = \min_{x \in \mathcal{X}} \left\{ f_n(x) = \frac{1}{n} \sum_{i=1}^n F(x, \xi_i) \right\}$$

Example

- Distributions of selling prices for each object is known
- Generate n scenarios by sampling selling prices for each object
- Solve each scenario independently
- Determine approximate optimal value and set of approximate optimal decisions from the n solved scenarios

Stochastic Decision Making III



Expected Value Variance Principle

- x is evaluated by a preference function using expected value and variance
- Incorporating variance for a more precise objective function
- Variance is weighted by q : $\mu(x) + q \cdot \sigma^2(x)$
 - $q = -1$ high affinity to risk
 - $q = 0$ neutral to risk
 - $q = 1$ risk-averse

Example

- Burglar knows distributions of selling prices for each object
 - E.g., through assumption based on historical data
- Calculates expected values (and variances)
- Chooses combination of objects with best total expected selling prices

Quasi-stochastic Decision Making I



Hurwicz-Criterion

- Consider worst and best realization
- Weighted sum with optimism-parameter $\lambda \in [0,1]$:
 $(1 - \lambda) \cdot \text{"worst case"} + \lambda \cdot \text{"best case"}$

Example

- Burglar knows (historical) selling prices for each objects
- Choose different scenarios
- Calculates weighted mean / sum prices for each object
- Chooses combination of objects with best total prices as determined

Quasi-stochastic Decision Making II



Minmax-Regret-Criterion

- Comparing a pessimistic decision in a certain scenario with best possible decision (assuming complete information) in the same scenario
- Regret: Deviation of the worst decision from the best possible decision in the same scenario

Example

- Regret of burglar not realizing the best case
- Burglar knows (historical) selling prices for each objects
- Calculate difference of best and worst cases for each combination
 - Worst selling price for taken objects
 - Best selling price for objects left behind
- Results in taking objects with a stable market price and leaving things behind that only yield a high price in the optimistic case.

Agenda

- Uncertainty
- Modeling Uncertainty (Descriptive and Predictive Analytics)
- Decision Making under Uncertainty (Prescriptive Analytics)

- **Case Study: Robust Vehicle Routing in Urban Logistics**

Case Study: Robust Vehicle Routing for Urban Logistics

Challenges of urban logistics

- High demand and cost-pressure
- **High Customer Expectation**



Urban Trafficsystems

- Limited infrastructure: Traffic congestion
- Volatile travel times: **Uncertainty**

Integration of Informationen on
Uncertainty of Travel Times
into Vehicle Routing for Urban Logistics

Groß, P.-O., Ulmer, M.W., Ehmke, J.F., Mattfeld, D.C., 2015. Exploiting Travel Time Information for Reliable Routing in City Logistics. Transportation Research Procedia 10, 652 – 661.

Groß, P.-O., Ehmke, J.F., Mattfeld, D.C., 2020. Interval travel times for robust synchronization in city logistics vehicle routing, Transportation Research Part E: Logistics and Transportation Review, Volume 143, 2020

Roles in Urban Logistics

Shipping Company

- Online retailer
- Number of known customers
- Commission logistic service provider with delivery of goods
- Demand α -service level, based on promised delivery dates

Logistic service provider

- Parcel Services
- Operate a fleet of delivery vehicles
- A-priori / “day-ahead” planning of deliveries
- Minimize travel time

α -service level is affected by travel times

→ Information w.r.t. uncertainty of travel times must be considered in the planning phase

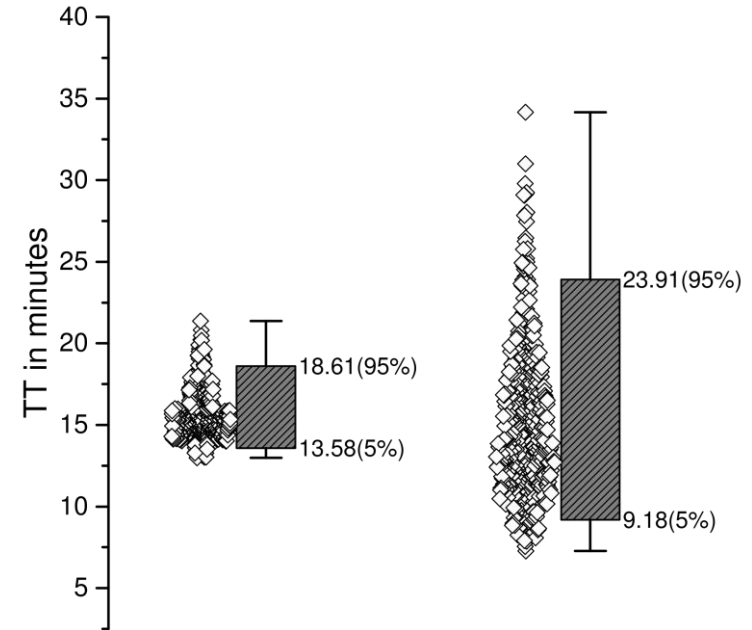
Modelling uncertain travel times

	Stochastic	Quasi-stochastic
Representation	Distribution of travel times with expected value μ and variance σ^2 [Lecluyse et al. (2009)]	Intervals of travel times with limits $[l, u]$ [Karasan et al. (2001)]
Data requirements	Large number of observations [Ehmke et al. (2016)]	Small number of observations [Groß et al. (2015)]
Modeling	Type of distribution / parametrization [Susilawati et al. (2013)]	Determining interval limits
Aggregation	Convolution [Gómez et. al. (2015)]	Interval arithmetics [Alefeld and Mayer (2000)]
Interpretability	Abstract / “Complex”	Intuitive / “Simple”

→ Interval travel times is an appropriate information model for urban logistics

Generating the information model data

- Shifted Gamma distribution
 - Parameter
 - Mean Travel Time
 - Coefficient of Variation
- Sampling Approach
 - Randomly draw 500 observations



Example of a connection. Parameters:
Mean travel time = 15.5
Coefficient of variance = 10% (left) and 30% (right)

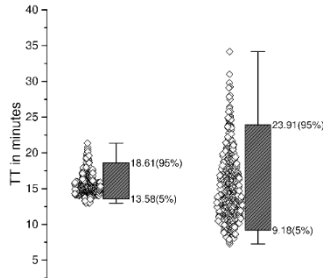
Quelle: Groß, P.-O., Ulmer, M.W., Ehmke, J.F., Mattfeld, D.C., 2015. Exploiting Travel Time Information for Reliable Routing in City Logistics. Transportation Research Procedia 10, 652 – 661.

Information Model

Two travel time matrices

- U : upper limit of interval travel times
- 95% Quantile (on peak traffic)
- L : lower limit of interval travel times
- 5% Quantile (off peak traffic)

$$U = \begin{pmatrix} u_{ij} & \dots \end{pmatrix}$$
$$L = \begin{pmatrix} l_{ij} & \dots \end{pmatrix}$$
$$l_{ij} \leq u_{ij}$$



Quelle: Groß, P.-O., Ulmer, M.W., Ehmke, J.F., Mattfeld, D.C., 2015. Exploiting Travel Time Information for Reliable Routing in City Logistics. Transportation Research Procedia 10, 652 – 661.

Robust TSP with Minmax Regret-Criterion

Cost of the candidate tour in scenario r:	14	20	18
Cost of the cost-minimal tour (TSP) in scenario r:	9	14	12
Regret of the candidate tour :	5	6	6
<u>Solution:</u>	Best minmax regret : 5 Robust tour : 1 → 2 → 3 → 4 → 1		

Legend: ——— Tour edges ——— Unused edges

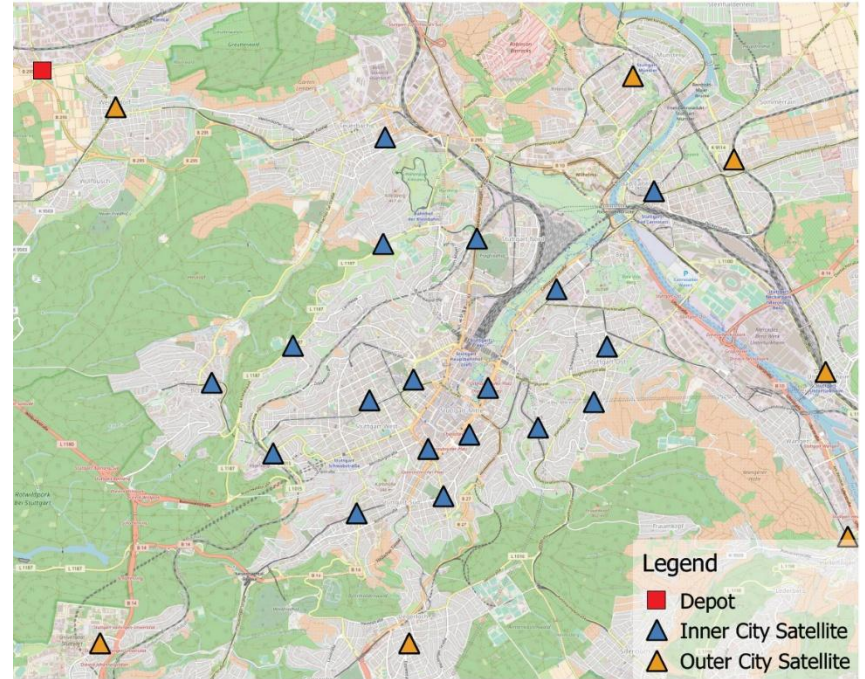
Quelle: Groß, P.-O., Ulmer, M.W., Ehmke, J.F., Mattfeld, D.C., 2015. Exploiting Travel Time Information for Reliable Routing in City Logistics. Transportation Research Procedia 10, 652 – 661.

What to achieve?

- We aim at finding a route that can be implemented for most traffic speed scenarios
- A tradeoff of efficiency in terms of travel time and service reliability in terms of punctuality
- Considering the best possible case is not an option at all. This will violate feasibility
- A mean of best and worst traffic speed per road segment may deteriorate service quality
- Considering worst case traffic speed leads to a poor efficiency, but good service quality
- Hopefully, a regret based approach balances efficiency and reliability
- Evaluation: Sum of earliness and lateness (=deviation) from self-imposed arrival times

Problem Setting

- “Two-Tier” Delivery network
- Satellites as transfer points
- Delivery from depot to transfer points using a fleet of vehicles
- Final delivery originates from each transfer point
- Objective: Minimize travel times including wait times at transfer points



Results

- Method:
 - VRP using minmax regret criterion
 - Metaheuristic
- Evaluation:
 - AVG (mean), UB (worst-case), ITT (interval)
- Evaluation criteria
 - Vehicles used
 - Total travel time in minutes
 - Mean deviation of arrival at satellite (desynchronized)

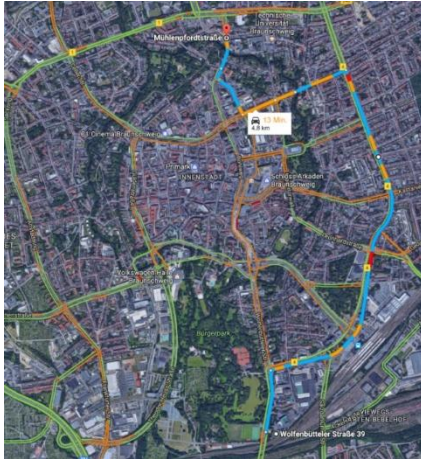
Traffic Situation	Travel Time Model	Vehicles Used	Avg. Duration (hh:mm)	Avg. Time Desynchronized (hh:mm)
On Peak	AVG	3	12:30	08:14
	UB	9	26:22	09:50
	ITT	5	16:05	02:39
Off Peak	AVG	3	11:30	05:16
	UB	6	17:44	05:02
	ITT	4	13:40	02:05

Summary

- Uncertainty should be considered in information and decision model
- Uncertainty affects many applications. Uncertainty may have significant affect on the decisions made.
- Modeling uncertainty appropriately is necessary to make sensible use of decision-relevant data.
- Decision models must be selected in dependence of application and derived modelling of uncertainty.

Further Questions

- Can we react to changes?
- Example: The TSP is re-planned based on current traffic information



→ Dynamic decision making

- Can we predict future uncertain developments?
- Example: Some regions with risk of traffic congestions can be avoided.



→ Anticipation

→ Stochastic dynamic decision making